

# CEDRIC MASTERS AUTOS WORKSHOP & DEALERSHIP MANAGEMENT SYSTEM

A single system to manage every vehicle that comes through the doors — from service intake to sale and handover. Built for the service bay, the showroom, and the back office.

|                              |                            |                              |                      |
|------------------------------|----------------------------|------------------------------|----------------------|
| DOCUMENT ID<br>CMA-SRS-001   | VERSION<br>v1.0.0          | STATUS<br>Approved           | DATE<br>07 Aug 2026  |
| PLATFORM<br>Web (SaaS-ready) | STACK<br>TanStack · Convex | FIRST SITE<br>Lagos, Nigeria | BUILD<br>MVP + P1-P4 |

## 01 DOCUMENT CONTROL

---

## REVISION HISTORY

| REV | DATE        | AUTHOR   | CHANGE                |
|-----|-------------|----------|-----------------------|
| 0.1 | 17 Jul 2026 | eng-team | Draft from PRD        |
| 0.2 | 18 Jul 2026 | eng-team | Add RBAC, audit trail |
| 1.0 | 07 Aug 2026 | eng-team | Baseline release      |

## APPROVALS

| ROLE             | OWNER          | SIGN-OFF |
|------------------|----------------|----------|
| Product Owner    | Cedric Masters | —        |
| Engineering Lead | eng-team       | —        |
| QA Lead          | qa-team        | —        |

Sign-off is recorded when each role acknowledges this baseline.

## TERMINOLOGY

- **Job** — a repair / service work order
- **CSR** — Customer Service Representative (service advisor)
- **Parts request** — technician stock request for a job
- **RBAC** — Role-Based Access Control

## REFERENCED DOCS

- PRD.md — product requirements
- ROADMAP.md — feature status
- docs/mock-accounts.md — demo users
- user\_manual.pdf — operator guide

## GLOSSARY

- **Kobo** — NGN subunit; all money stored as integer kobo
- **VAT** — Value-Added Tax (NGN default 7.5%)
- **OEM** — Original Equipment Manufacturer

## CONVENTIONS

- Priority: **P0** core · **P1** perf · **P2+** later
- Requirement IDs are stable trace keys

# 02 INTRODUCTION

## PURPOSE

This document defines the functional and non-functional requirements for the **Cedric Masters Autos** workshop and dealership management system. It is the baseline agreement between product, engineering, and quality — the single reference for what the system does, how it behaves, and how it will be verified.

## SCOPE

### # IN SCOPE — MVP

- After-sales service: customers, check-in, technician intake, parts catalogue, labour rates, invoicing, payments, 8-status job workflow
- Vehicle sales: inventory, leads/CRM, sales orders, delivery handover
- Administration: 7 roles, user management, VAT & labour settings
- Cross-cutting: immutable audit trail, role-based access control, Excel import of parts

### # OUT OF SCOPE — POST-MVP

- Customer self-service portal (history, downloads, online booking)
- Reporting & dashboards (revenue, productivity, parts usage)
- Multi-branch / multi-tenant organisation scoping
- Trade-in management & sales commission tracking
- Accounting integration (QuickBooks, Sage) & Excel export
- Email/SMS delivery of reset codes & invoices

## DEFINITIONS & ABBREVIATIONS

| TERM          | MEANING                                                                         |
|---------------|---------------------------------------------------------------------------------|
| Job           | Repair or service work order bound to one customer vehicle                      |
| Check-in      | The walk-in intake flow that creates customer → vehicle → job                   |
| Parts request | Technician stock request linked to a job; approved and dispatched by Inventory  |
| Job item      | Billable line — either a dispatched part or a labour line                       |
| RBAC          | Role-Based Access Control; enforced server-side on every function               |
| Kobo          | Integer subunit of the Naira (100 kobo = 1 NGN); money is never stored as float |
| CSR / Advisor | Front-desk service advisor; owns check-in and customer registration             |

03

OVERALL DESCRIPTION

PRODUCT PERSPECTIVE

Cedric Masters Autos is a realtime, serverless web application. The frontend is **TanStack Start (React 19)** with **TanStack Router** and **Tailwind v4**; the backend is **Convex** (typed, reactive database + functions). Data flows one direction: the browser subscribes to typed queries, and every write is a validated, authorized, audited mutation. Roles are enforced in the backend, never merely hidden in the UI.

USER ROLES & OPERATING ENVIRONMENT

THE SEVEN ROLES

| ROLE              | CODE             | PRIMARY DUTIES                           |
|-------------------|------------------|------------------------------------------|
| Admin             | admin            | Full access, user management, rates      |
| Service Advisor   | csr              | Check-in, customers, appointments        |
| Technician        | technician       | Diagnose, work jobs, request parts       |
| Inventory Manager | inventoryManager | Parts catalogue, stock, requests         |
| Finance           | finance          | Invoices, approval, payments             |
| Manager           | manager          | Approve invoices, complete jobs, oversee |
| Sales Rep         | salesRep         | Leads, sales orders, deliveries          |

ENVIRONMENT & CONSTRAINTS

- Modern evergreen browsers; responsive down to mobile
- Deployment: Vercel (Nitro adapter, .output/), Convex cloud backend
- Auth: Convex Auth, password provider (email + password)
- Validation: Zod v4 shared between client and server
- Money: integer kobo everywhere; VAT rounded to kobo
- TypeScript 7 (tsgo) — typecheck ~2 s

ASSUMPTIONS & DEPENDENCIES

## ASSUMPTIONS

Staff are named users with assigned roles; one person may hold one role at a time; NGN is the transactional currency in v1.

## DEPENDENCIES

Convex availability, Google Fonts for rendering, browser print support for job cards and invoices.

## OPEN QUESTIONS

Discounts & limits, VAT exemptions, corporate net-30 credit terms, first-time vs returning customer flows.

# 04 FUNCTIONAL REQUIREMENTS

## SERVICE MODULE — AFTER-SALES

### FR-SVC-01 · CUSTOMER DATABASE

|              |                                                                                                                                                           |       |                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------|
| Priority     | P0                                                                                                                                                        | Actor | CSR · Manager · Admin |
| Requirement  | The system SHALL maintain a central customer record (name, phone, optional email, address) and SHALL support create, update, and search by name or phone. |       |                       |
| Verification | Unit + E2E; manual search smoke test (convex/customers.ts, search indexes on name & phone)                                                                |       |                       |

### FR-SVC-02 · ONE-STOP CREATION

|             |                                                                                                                      |       |                       |
|-------------|----------------------------------------------------------------------------------------------------------------------|-------|-----------------------|
| Priority    | P0                                                                                                                   | Actor | CSR · Manager · Admin |
| Requirement | The system SHALL allow a customer, their vehicle, and a job to be created in a single flow without leaving the form. |       |                       |

### FR-SVC-03 · REPAIR HISTORY

|             |                                                                                                                                        |       |                                 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------|
| Priority    | P0                                                                                                                                     | Actor | CSR · Finance · Manager · Admin |
| Requirement | The system SHALL show the full repair history for a customer (all jobs, vehicles, invoices, and statuses) on the customer detail page. |       |                                 |

### FR-SVC-04 · WALK-IN CHECK-IN

|             |                                                                                                                                                                                        |       |                       |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------|
| Priority    | P0                                                                                                                                                                                     | Actor | CSR · Manager · Admin |
| Requirement | The system SHALL provide a 3-step check-in flow: locate/create the customer, locate/create the vehicle, capture the complaint and create the job in checkedIn status with a timestamp. |       |                       |

## FR-SVC-05 · APPOINTMENT BOOKING

|             |                                                                                                                                                                                                                                               |       |                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------|
| Priority    | P0                                                                                                                                                                                                                                            | Actor | CSR · Manager · Admin |
| Requirement | The system SHALL support booking appointments with Today/Week/Month range views, SHALL reject past dates (server + client), and SHALL require vehicle make, model, plate, and complaint. Appointments move scheduled → checkedIn → cancelled. |       |                       |

## FR-SVC-06 · TECHNICIAN JOB INTAKE

|             |                                                                                                                                                                                                                  |       |                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------|
| Priority    | P0                                                                                                                                                                                                               | Actor | Technician · Manager · Admin |
| Requirement | The system SHALL allow assignment of a technician, recording of a diagnosis, and transition of the job through the state machine. A technician SHALL only act on jobs assigned to them (admin/manager excepted). |       |                              |

## FR-SVC-07 · SPARE PARTS CATALOGUE

|             |                                                                                                                                                                                                                                                                |       |                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------|
| Priority    | P0                                                                                                                                                                                                                                                             | Actor | Inventory Manager · Manager · Admin |
| Requirement | The system SHALL support CRUD on parts (unique code, description, cost & selling price, stock quantity, reorder level), one-click CSV import, a low-stock badge when stockQty ≤ reorderLevel, and an immutable stock-movement audit trail (in / out / adjust). |       |                                     |

## FR-SVC-08 · LABOUR RATE CONFIGURATION

|             |                                                                                                                                        |       |                           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------|
| Priority    | P0                                                                                                                                     | Actor | Finance · Manager · Admin |
| Requirement | The system SHALL maintain a lookup of job types to fixed prices (e.g., "Oil change = NGN 5,000") independent of technician or vehicle. |       |                           |

## FR-SVC-09 · INVOICE GENERATION

|             |                                                                                                                                                                                                                                                                                                                |       |                           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------|
| Priority    | P0                                                                                                                                                                                                                                                                                                             | Actor | Finance · Manager · Admin |
| Requirement | The system SHALL compute invoices from part + labour line items with a configurable VAT rate (default 7.5%), snapshot line items, support regeneration (re-sync) until paid, and require manager approval before the customer-facing copy is valid. Regeneration resets approval; a paid invoice is immutable. |       |                           |

## FR-SVC-10 · PAYMENT RECORDING

|             |                                                                                                                                                                                                |       |                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------|
| Priority    | P0                                                                                                                                                                                             | Actor | Finance · Manager · Admin |
| Requirement | The system SHALL record payments by method (cash / transfer / card), support partial payments with running balance, and auto-mark the invoice paid when <code>amountPaid ≥ grandTotal</code> . |       |                           |

## FR-SVC-11 · JOB PROGRESS (8-STATUS STATE MACHINE)

|              |                                                                                                                                                                                                                                                                                                                     |       |                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------|
| Priority     | P0                                                                                                                                                                                                                                                                                                                  | Actor | All staff per transition |
| Requirement  | The system SHALL model jobs as: <code>checkedIn → assigned → diagnosed → inProgress → readyForPickup → completed → paid</code> , with <code>waitingRelease</code> as a holding state when parts are needed. Each transition SHALL be guarded by <code>canTransition</code> and SHALL record a per-status timestamp. |       |                          |
| Verification | tests/unit/job-utils.test.ts — 5 suites, full transition matrix                                                                                                                                                                                                                                                     |       |                          |

## FR-SVC-12 · PARTS REQUEST WORKFLOW

|             |                                                                                                                                                                                                                                                                                                                                                                                   |       |                                  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------|
| Priority    | P0                                                                                                                                                                                                                                                                                                                                                                                | Actor | Technician → Inventory → Finance |
| Requirement | The system SHALL move a job to <code>waitingRelease</code> when a technician requests parts, SHALL let Inventory approve/reject/dispatch, SHALL auto-decrement stock on dispatch, add part line items to the job, and re-sync the invoice. Reversal SHALL restore stock, remove line items, and re-sync the invoice. Work SHALL NOT start while approved/pending requests remain. |       |                                  |

## FR-SVC-13 / 14 · PRINTABLE JOB CARD & INVOICE

|             |                                                                                                                                                             |       |                                              |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------|
| Priority    | P0                                                                                                                                                          | Actor | Manager · Technician · CSR · Admin / Finance |
| Requirement | The system SHALL render A4-printable job cards and invoices populated from backend data, hidden on screen and revealed only for print (hidden print:block). |       |                                              |

### SALES MODULE

#### FR-SLS-01 · VEHICLE INVENTORY

|          |                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority | P0                                                                                                                                                                                     |
| Actor    | Sales Rep · CSR · Manager · Admin                                                                                                                                                      |
| Req      | CRUD vehicles (make, model, year, colour, VIN, plate, cost, selling price) with status inStock / reserved / sold / customerOwned, stock quantity, reorder level, and stock adjustment. |

#### FR-SLS-02 · CUSTOMER LEADS & CRM

|          |                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------|
| Priority | P0                                                                                                                       |
| Actor    | Sales Rep · CSR · Manager · Admin                                                                                        |
| Req      | CRUD leads with stages new → contacted → qualified → won / lost, follow-up notes timeline, and next-follow-up reminders. |

#### FR-SLS-03 · SALES ORDERS

|          |                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority | P0                                                                                                                                                   |
| Actor    | Sales Rep · CSR · Manager · Admin                                                                                                                    |
| Req      | Create order (agreed price + deposit), auto-reserve vehicle, track balance & payment log, complete only when balance is zero, cancel restores stock. |

#### FR-SLS-04 · DELIVERY HANDOVER

|          |                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------|
| Priority | P0                                                                                                             |
| Actor    | Sales Rep · CSR · Manager · Admin                                                                              |
| Req      | Record delivery checklist (keys, manual, toolkit, inspection) against the sales rep with a handover timestamp. |

### ADMINISTRATION

FR-ADM-01 · USER ROLES & MANAGEMENT

|          |                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority | P0                                                                                                                                                                                                                |
| Actor    | Admin                                                                                                                                                                                                             |
| Req      | Admin SHALL list users, assign any of the 7 roles, activate/deactivate accounts. Deactivated users SHALL fail server-side auth checks. A one-time bootstrap promotes the first admin only when zero admins exist. |

FR-ADM-02/03 · SETTINGS

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| Priority | P0                                                                                  |
| Actor    | Finance · Manager · Admin                                                           |
| Req      | Configure the VAT rate (default 7.5%) and manage labour types (name + fixed price). |

CROSS-CUTTING

FR-XCT-01 · AUDIT TRAIL

Every mutation SHALL append an immutable audit record: user, action, entity, entity id, timestamp (convex/lib/audit.ts).

FR-XCT-02 · RBAC

Server SHALL default-deny: a null role is never authorized, admin bypasses, every mutation guards its roles via requireRole. UI reflects the same matrix.

FR-XCT-05 · RATE LIMITING

Targeted fixed-window limiting on financial and write-heavy mutations is **proposed, not yet implemented** — see the [Throttling Plan](#).

## 05

## NON-FUNCTIONAL REQUIREMENTS

| CATEGORY        | ID         | REQUIREMENT                                                                                                                                   |
|-----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Performance     | NFR-PER-01 | Time-to-first-paint for authenticated routes $\leq 2$ s on a normal office connection; typecheck $\leq 5$ s (currently $\sim 2$ s with tsgo). |
| Performance     | NFR-PER-02 | Read queries resolve $\leq 500$ ms for catalogues $\leq 10,000$ rows under normal load.                                                       |
| Scalability     | NFR-SCL-01 | Support $\geq 50$ concurrent staff sessions per site; Convex-backed realtime subscriptions without client pooling.                            |
| Availability    | NFR-AVA-01 | 99.5% availability for the hosted app; offline edits out of scope for v1.                                                                     |
| Security        | NFR-SEC-01 | All writes authorized server-side; default-deny on unknown roles; sensitive data not exposed to unauthorized roles.                           |
| Usability       | NFR-US-01  | Core flows (check-in, invoice, order) completable in $\leq 3$ min by a trained CSR; responsive down to 360 px.                                |
| Reliability     | NFR-RLB-01 | Monetary integrity is invariant: integer kobo arithmetic, VAT rounding to kobo, balance never negative.                                       |
| Maintainability | NFR-MNT-01 | Shared typed modules (enums, schemas, job-utils) as single sources of truth; zero hardcoded colours in components.                            |
| Portability     | NFR-PRT-01 | Deploys to Vercel via Nitro .output/; runs with pnpm dev locally against a Convex dev backend.                                                |
| Compliance      | NFR-CMP-01 | Tax-inclusive pricing per configurable NGN VAT; audit trail retained for review (retention per site policy).                                  |

## USER INTERFACES

| ROUTE                                                      | PURPOSE                      |
|------------------------------------------------------------|------------------------------|
| /                                                          | Dashboard (role-aware)       |
| /auth/login · reset-password                               | Auth                         |
| /service/checkin                                           | 3-step walk-in               |
| /service/customers · customer/\$id                         | Customers + history          |
| /service/jobs · job/\$id                                   | Job board + detail           |
| /service/appointments                                      | Booking range view           |
| /service/parts · vehicles                                  | Catalogue / vehicles         |
| /service/finance                                           | Invoices, payments, settings |
| /sales/inventory · leads · orders · lead/\$id · order/\$id | Sales pipeline               |
| /admin/users                                               | User management              |

## SOFTWARE &amp; HARDWARE INTERFACES

- **Convex backend** — typed queries/mutations via generated api module; reactive subscriptions via `@convex-dev/react-query`
- **Convex Auth** — password provider; session via `useConvexAuth`
- **Auth HTTP routes** — `convex/http.ts` mounts auth endpoints
- **CSV import** — parts bulk load via mutation
- **Printer** — browser print pipeline for job card / invoice (A4)
- **Email** — reset-password provider currently logs codes to console (see **SECURITY NOTE**); SMTP provider is a planned interface

07

PERFORMANCE BENCHMARKS

Targets below are baselines to hold the MVP at; the P1 track (Vite warmup, single-auth-roundtrip gating, query guards, lazy devtools, workers=2 Playwright) already lands several of them.

MEASURED TODAY

| METRIC                  | BASELINE      |
|-------------------------|---------------|
| Typecheck (TS 7 / tsgo) | ~2 s          |
| Guest redirect to login | 1 roundtrip   |
| E2E workers             | 2 (8 GB host) |
| E2E suite               | 6/6 green     |

TARGETS

| METRIC                     | TARGET   |
|----------------------------|----------|
| First paint (authed route) | ≤ 2 s    |
| Catalogue read             | ≤ 500 ms |
| Write (check-in / invoice) | ≤ 400 ms |
| Concurrent staff           | ≥ 50     |

# 08

## SECURITY REQUIREMENTS

### CONTROLS IN PLACE

- **Authentication** — email + password via Convex Auth; reset-password route
- **Authorization** — requireUser / requireRole on every protected function; default-deny
- **Data isolation** — technicians scoped to assigned jobs; invoices hidden from technicians
- **Validation** — Zod v4 on every write, shared client/server
- **Monetary integrity** — integer kobo, no floats
- **Auditability** — immutable audit log on all mutations
- **State integrity** — transition guards + invoice immutability after payment

### OPEN RISKS & MITIGATIONS

| RISK                                                      | SEVERITY | MITIGATION                                                                |
|-----------------------------------------------------------|----------|---------------------------------------------------------------------------|
| Reset tokens logged to console (convex/auth.ts)           | HIGH     | Swap console log for SMTP/SendGrid delivery before any production rollout |
| Overpayment beyond grandTotal accepted by payments.record | MED      | Cap payment amount at remaining balance (server)                          |
| No request rate limiting                                  | MED      | Adopt <a href="#">Throttling Plan</a> (proposed)                          |
| bootstrapFirstAdmin race at first sign-up                 | MED      | Guard verified by adminExists; audit logged; restrict at deploy gate      |

09

VERIFICATION METHODS

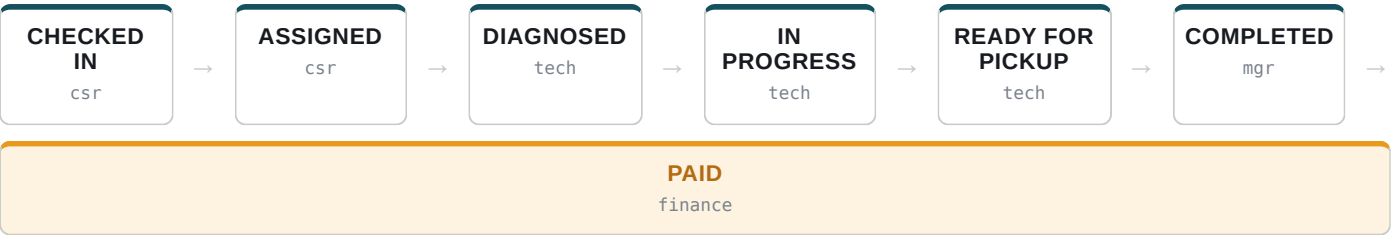
| LAYER  | TOOLING                | COVERS                                                                                  |
|--------|------------------------|-----------------------------------------------------------------------------------------|
| Unit   | Vitest                 | Job state machine, invoice totals, auth-utils, schemas, formatting (tests/unit/)        |
| Authz  | Vitest vs live Convex  | Anonymous rejection on queries/mutations; auto-skips without CONVEX_URL (tests/convex/) |
| E2E    | Playwright (2 workers) | Auth/nav flows (6/6 green), jobs CRUD, screenshots; hydration-robust with networkidle   |
| Manual | Smoke script           | All 7 role accounts; check-in → parts request → dispatch → invoice → approve → pay path |
| Print  | Browser print          | Job card & invoice A4 rendering from live data                                          |

ACCEPTANCE GATE

Release requires: unit + authz suites green, E2E auth/nav green, one end-to-end paid-job walkthrough on seeded data, and a printed job card + invoice matching the live record.

# 10 USER WORKFLOWS

## W1 · JOB LIFECYCLE — 8-STATUS STATE MACHINE



Transitions are validated by `canTransition` in `src/lib/job-utils.ts`; role gates live in `convex/jobs.ts`. Each transition stamps its own timestamp field.

## W2 · WALK-IN CHECK-IN (3-STEP)

**01** CSR  
**Locate or create the customer**  
Search by name or phone; returning customers auto-populate.

**02** CSR  
**Locate or create the vehicle**  
Plate lookup; duplicate plates rejected server-side.

**03** CSR  
**Capture complaint & check in**  
Creates job in `checkedIn` with `checkInTs` and audits the action.

## W3 · PARTS REQUEST → DISPATCH

01

TECHNICIAN

**Request parts on an assigned job**

Stock is validated at request time; the job moves to waitingRelease.

02

INVENTORY MANAGER

**Review request**

Approve or reject (job returns to diagnosed if rejected).

03

INVENTORY MANAGER

**Dispatch**

Stock decremented, movements logged, part line items added, invoice re-synced, job resumes.

04

INVENTORY MANAGER

**Reverse (correction)**

Stock restored, items removed, invoice re-synced.

W4 · INVOICE → PAYMENT

01

FINANCE

**Generate / regenerate invoice**

Snapshot of part + labour items; VAT applied at configured rate.

02

MANAGER

**Approve invoice**

Required before payments or markPaid are accepted.

03

FINANCE

**Record payments**

Cash / transfer / card; partial payments track balance.

04

FINANCE

**Mark paid**

Auto-flagged when fully settled; invoice becomes immutable.

W5 · SALES ORDER → HANDOVER

01

SALES REP

**Create order from a lead**

Agreed price + deposit; vehicle stock decremented and reserved when zero.

---

02

SALES REP

**Record payments**

Balance tracked; payments capped at remaining balance.

---

03

SALES REP

**Complete order**

Only when balance is zero; vehicle marked sold.

---

04

SALES REP

**Delivery handover**

Checklist: keys, manual, toolkit, inspection — recorded against the rep.

## 11

## BUSINESS RULES &amp; DATA REQUIREMENTS

## BUSINESS RULES

| RULE  | STATEMENT                                                                                     | REFERENCE                |
|-------|-----------------------------------------------------------------------------------------------|--------------------------|
| BR-01 | All money stored as integer kobo (100 kobo = NGN 1); no floats.                               | schemas/common.ts        |
| BR-02 | VAT applies at a configurable rate, default 7.5%, rounded to kobo.                            | settings.ts · invoice.ts |
| BR-03 | Invoice line items are a snapshot; regeneration resets approval; paid invoices are immutable. | invoices.ts              |
| BR-04 | Job transitions follow canTransition; illegal transitions are rejected server-side.           | job-utils.ts             |
| BR-05 | Only Manager and Admin may complete a job.                                                    | jobs.complete            |
| BR-06 | A job is marked paid only when the invoice is approved and fully settled.                     | jobs.markPaid            |
| BR-07 | Part line items enter a job only via Inventory dispatch; manual part adds are blocked.        | jobs.addJobItem          |
| BR-08 | Dispatch decrements stock; reversal restores stock and removes line items.                    | partsRequests.ts         |
| BR-09 | Work may not start while an approved/pending parts request exists.                            | jobs.startWork           |
| BR-10 | Sales orders require stock; completing requires zero balance; cancelling restores stock.      | salesOrders.ts           |
| BR-11 | Appointments must be in the future; make/model/plate/complaint are mandatory.                 | appointments.create      |

| RULE  | STATEMENT                                                                | REFERENCE                              |
|-------|--------------------------------------------------------------------------|----------------------------------------|
| BR-12 | Technicians act only on jobs assigned to them (admin/manager excepted).  | <code>jobs.ts</code>                   |
| BR-13 | A part is low-stock when $\text{stockQty} \leq \text{reorderLevel}$ .    | <code>parts.lowStock</code>            |
| BR-14 | Vehicle plates are unique (normalized lowercase) at creation.            | <code>vehicles.create</code>           |
| BR-15 | Deactivated users fail all server-side checks.                           | <code>lib/auth.ts</code>               |
| BR-16 | Bootstrap admin promotion succeeds only when zero admins exist.          | <code>users.bootstrapFirstAdmin</code> |
| BR-17 | Every mutation records an audit entry (user, action, entity, timestamp). | <code>lib/audit.ts</code>              |
| BR-18 | Sales-order payments may not exceed the remaining balance.               | <code>salesOrders.addPayment</code>    |

## DATA REQUIREMENTS

---

## ENTITY GROUPS

| TABLE     | KEY FIELDS                                    | INDEXES                            |
|-----------|-----------------------------------------------|------------------------------------|
| customers | name,<br>phone,<br>email,<br>address          | phone · name s                     |
| vehicles  | make,<br>model, plate,<br>status,<br>stockQty | status · owner · plate s           |
| jobs      | status,<br>complaint, 8<br>ts fields          | status · technicianId · customerId |
| parts     | code, prices,<br>stockQty,<br>reorderLevel    | code · search                      |
| invoices  | lineItems,<br>totals,<br>approved,<br>paid    | jobId                              |
| auditLogs | user, action,<br>entity, ts                   | entityId                           |

## INTEGRITY & RETENTION

- Referential integrity via typed ids (v.id) across jobs → vehicles/customers/users
- Stock movements are append-only; part quantity is derived state
- Audit log is append-only; no delete mutation exists
- Seed data for demo: 45 parts, 18 customers, 19 vehicles, 7 jobs, 3 invoices, 3 leads, 2 sales orders
- Retention: audit log and payments retained per site policy (backup + review); no auto-purge in v1

# 12 TRACEABILITY MATRIX

| REQUIREMENT            | VERIFICATION                   | STATUS   |
|------------------------|--------------------------------|----------|
| FR-SVC-01..14          | Unit + E2E + manual smoke      | VERIFIED |
| FR-SLS-01..04          | E2E + manual (sales module)    | PARTIAL  |
| FR-ADM-01..03          | Manual role walkthrough        | VERIFIED |
| FR-XCT-01/02           | Authz tests + audit assertions | VERIFIED |
| FR-XCT-05 (throttling) | Planned — see throttling plan  | PROPOSED |
| NFR-PER-*              | Measured baselines             | PARTIAL  |
| NFR-RLB-01             | Invoice unit tests             | VERIFIED |